

There is a natural projection $p: O(n) \rightarrow V_{n,k}$ sending $\alpha \in O(n)$ to the k -frame consisting of the last k columns of α , which are the images under α of the last k standard basis vectors in $\mathbb{R}^n$. This projection is onto, and the preimages of points are precisely the cosets $\alpha O(n-k)$, where we embed $O(n-k)$ in $O(n)$ as the orthogonal transformations of the first $n-k$ coordinates of $\mathbb{R}^n$. Thus $V_{n,k}$ can be viewed as the space $O(n)/O(n-k)$ of such cosets, with the quotient topology from $O(n)$. This is the same as the previously defined topology on $V_{n,k}$ since the projection $O(n) \rightarrow V_{n,k}$ is a surjection of compact Hausdorff spaces.

When $k < n$ the projection $p: SO(n) \rightarrow V_{n,k}$ is surjective, and $V_{n,k}$ can also be viewed as the coset space $SO(n)/SO(n-k)$. We can use this to induce a CW structure on $V_{n,k}$ from the CW structure on $SO(n)$. The cells are the sets of cosets of the form $e^J SO(n-k) = e^{i_1} \cdots e^{i_m} SO(n-k)$ for $n > i_1 > \cdots > i_m \geq n-k$, together with the coset $SO(n-k)$ itself as a 0-cell of $V_{n,k}$. These sets of cosets are unions of cells of $SO(n)$ since $SO(n-k)$ consists of the cells $e^J = e^{j_1} \cdots e^{j_\ell}$ with $n-k > j_1 > \cdots > j_\ell$. This implies that $V_{n,k}$ is the disjoint union of its cells, and the boundary of each cell is contained in cells of lower dimension, so we do have a CW structure.

Since the projection $SO(n) \rightarrow V_{n,k}$ is a cellular map, the structure of the cellular chain complex of $V_{n,k}$ can easily be deduced from that of $SO(n)$. For example, the cellular chain complex of $V_{2k+1,2}$ is just the complex C^{2k} defined earlier, while for $V_{2k,2}$ the cellular boundary maps are all trivial. Hence the nonzero homology groups of $V_{n,2}$ are

$$H_i(V_{2k+1,2}; \mathbb{Z}) = \begin{cases} \mathbb{Z} & \text{for } i = 0, 4k-1 \\ \mathbb{Z}_2 & \text{for } i = 2k-1 \end{cases}$$

$$H_i(V_{2k,2}; \mathbb{Z}) = \mathbb{Z} \quad \text{for } i = 0, 2k-2, 2k-1, 4k-3$$

Thus $SO(n)$ has the same homology and cohomology groups as the product space $V_{3,2} \times V_{5,2} \times \cdots \times V_{2k+1,2}$ when $n = 2k+1$, or as $V_{3,2} \times V_{5,2} \times \cdots \times V_{2k+1,2} \times S^{2k+1}$ when $n = 2k+2$. However, our calculations show that $SO(n)$ is distinguished from these products by its cup product structure with $\mathbb{Z}_2$ coefficients, at least when $n \geq 5$, since β_1^4 is nonzero in $H^4(SO(n); \mathbb{Z}_2)$ if $n \geq 5$, while for the product spaces the nontrivial element of $H^1(-; \mathbb{Z}_2)$ must lie in the factor $V_{3,2}$, and $H^4(V_{3,2}; \mathbb{Z}_2) = 0$. When $n = 4$ we have $SO(4)$ homeomorphic to $SO(3) \times S^3 = V_{3,2} \times S^3$ as we noted at the beginning of this section. Also $SO(3) = V_{3,2}$ and $SO(2) = S^1$.

Exercises

1. Show that a topological group that has a CW structure is an orientable manifold. [Consider the homeomorphisms $x \mapsto xg$ for a fixed group element g .]
2. Using the CW structure on $SO(n)$, show that $\pi_1 SO(n) \approx \mathbb{Z}_2$ for $n \geq 3$. Find a loop representing a generator, and describe how twice this loop is nullhomotopic.
3. Compute the Pontryagin ring structure in $H_*(SO(5); \mathbb{Z})$.

3.E Bockstein Homomorphisms

Homology and cohomology with coefficients in a field, particularly $\mathbb{Z}_p$ with p prime, often have more structure and are easier to compute than with $\mathbb{Z}$ coefficients. Of course, passing from $\mathbb{Z}$ to $\mathbb{Z}_p$ coefficients can involve a certain loss of information, a blurring of finer distinctions. For example, a $\mathbb{Z}_{p^n}$ in integer homology becomes a pair of $\mathbb{Z}_p$'s in $\mathbb{Z}_p$ homology or cohomology, so the exponent n is lost with $\mathbb{Z}_p$ coefficients. In this section we introduce Bockstein homomorphisms, which in many interesting cases allow one to recover $\mathbb{Z}$ coefficient information from $\mathbb{Z}_p$ coefficients. Bockstein homomorphisms also provide a small piece of extra internal structure to $\mathbb{Z}_p$ homology or cohomology itself, which can be quite useful.

We will concentrate on cohomology in order to have cup products available, but the basic constructions work equally well for homology. If we take a short exact sequence $0 \rightarrow G \rightarrow H \rightarrow K \rightarrow 0$ of abelian groups and apply the covariant functor $\text{Hom}(C_n(X), -)$, we obtain

$$0 \rightarrow C^n(X; G) \rightarrow C^n(X; H) \rightarrow C^n(X; K) \rightarrow 0$$

which is exact since $C_n(X)$ is free. Letting n vary, we have a short exact sequence of chain complexes, so there is an associated long exact sequence

$$\cdots \rightarrow H^n(X; G) \rightarrow H^n(X; H) \rightarrow H^n(X; K) \rightarrow H^{n+1}(X; G) \rightarrow \cdots$$

whose ‘boundary’ map $H^n(X; K) \rightarrow H^{n+1}(X; G)$ is called a **Bockstein homomorphism**.

We shall be interested primarily in the Bockstein $\beta: H^n(X; \mathbb{Z}_m) \rightarrow H^{n+1}(X; \mathbb{Z}_m)$ associated to the coefficient sequence $0 \rightarrow \mathbb{Z}_m \xrightarrow{m} \mathbb{Z}_{m^2} \rightarrow \mathbb{Z}_m \rightarrow 0$, especially when m is prime, but for the moment we do not need this assumption. Closely related to β is the Bockstein $\tilde{\beta}: H^n(X; \mathbb{Z}_m) \rightarrow H^{n+1}(X; \mathbb{Z})$ associated to $0 \rightarrow \mathbb{Z} \xrightarrow{m} \mathbb{Z} \rightarrow \mathbb{Z}_m \rightarrow 0$. From the natural map of the latter short exact sequence onto the former one, we obtain the relationship $\beta = \rho \tilde{\beta}$ where $\rho: H^*(X; \mathbb{Z}) \rightarrow H^*(X; \mathbb{Z}_m)$ is the homomorphism induced by the map $\mathbb{Z} \rightarrow \mathbb{Z}_m$ reducing coefficients mod m . Thus we have a commutative triangle in the following diagram, whose upper row is the exact sequence containing $\tilde{\beta}$.

$$\begin{array}{ccccccc} H^n(X; \mathbb{Z}) & \xrightarrow{\rho} & H^n(X; \mathbb{Z}_m) & \xrightarrow{\tilde{\beta}} & H^{n+1}(X; \mathbb{Z}) & \xrightarrow{m} & H^{n+1}(X; \mathbb{Z}) \\ & & & \searrow \beta & \downarrow \rho & & \\ & & & & H^{n+1}(X; \mathbb{Z}_m) & & \end{array}$$

Example 3E.1. Let X be a $K(\mathbb{Z}_m, 1)$, for example $\mathbb{RP}^\infty$ when $m = 2$ or an infinite-dimensional lens space with fundamental group $\mathbb{Z}_m$ for arbitrary m . From the homology calculations in Examples 2.42 and 2.43 together with the universal coefficient theorem or cellular cohomology we have $H^n(X; \mathbb{Z}_m) \approx \mathbb{Z}_m$ for all n . Let us show that $\beta: H^n(X; \mathbb{Z}_m) \rightarrow H^{n+1}(X; \mathbb{Z}_m)$ is an isomorphism for n odd and zero for n even. If n is odd the vertical map ρ in the diagram above is surjective for $X = K(\mathbb{Z}_m, 1)$, as